Homework 3 - From a Research Question to Reusable Data

Purpose:

Goal: Develop a defensible biomedical research question and document the data needed to answer it in a way another researcher could understand and reuse.

Part 1 — Research question (15 points)

- Choose a biomedical topic.
- Write one clear research question.
- Identify the population, main predictor/exposure, and outcome.

Part 2 — FINER assessment (25 points)

Evaluate your question using each FINER criterion. Write 1–3 sentences for each criterion. If your evaluation reveals a weakness, revise the research question and show the revised version.

Criterion	Your assessment
Feasible	
Interesting	
Novel	
Ethical	
Relevant	

Part 3 — Data and FAIR (25 points)

Describe the data you would need. Then explain, in your own words, how each FAIR principle would apply to those data. “Accessible” does not mean that sensitive data must be publicly downloadable; describe appropriate access conditions.

FAIR principle	How it would apply
Findable	
Accessible	
Interoperable	
Reusable	

Part 4 — Documentation and provenance (35 points)

- Create a data dictionary for 6–8 variables. Include variable name, definition, type, units/coding, and missing-value rule.
- Using the supplied provenance scenario, document how clinical_data_clean.csv was produced from clinical_data_raw.csv.
- Complete the supplied README template so another researcher can understand the dataset and processing workflow.
- Briefly justify at least two cleaning/derivation decisions.

Submission

- One DOCX, Markdown or PDF containing Parts 1–3 and your provenance explanation.
- One completed CSV data dictionary.
- One completed README.md.
- Estimated effort: approximately 60–90 minutes.

Synthesis question

Suppose another researcher receives clinical_data_clean.csv but no data dictionary, README, or processing history. What problems could occur, and which FAIR principles would be most affected?